

SQL Planning

Road Entity

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Business rules

Business Rules:(Each ENTITY_1 May/Must Relationship_Verb_Phrase number ENTITY_2)

1. Road → Category

- Each Road must be assigned to one or many Categories

Category → Road

- Each Category may have zero or many Road.

2. Road → Location

- Each Road must have one or many Locations.

Location → Road

- Each Location may have zero or many Road.

3. Road → Projects (Some roads will not have project)

- Each Road may have zero or many Project

Project → Road

- Each Project must be assigned to one or many Road

4. Road → Road (Sub-Section) (Dirt road or Queenstreet)

- Each Road may have zero or many Road.

Road → Road

- Each Road(Sub-Section) maybe a subsection of only one Road.

5. Project → Contracts

- Each Project must have one or many Contracts.

Contracts → Project

- Each Contracts may be assigned zero or many project

6. Project → Staff

- Each Project must have one or many Staff

Staff —> Project

- Each Staff may have zero or many Project.

7. Staff —> Role

- Each Staff must be assigned to one or many Roles.

Role —> Staff

- Each Role must have one or many Staff.

8. Contract —> Contractor

- Each Contract must have only one Contractor

Contractor —> Contract

- Each Contractor may have zero or many Contracts

9. ** Contract —> Role ******

- Each Contract must have one or many role

Role —> Contract

- Each Role must have one or many contracts.

Planning ERD attributes

Planning

Road Entity:

- Road ID (Primary key)
- Road name
- Description
- Road Length
- Category (Foreign Key)
- Sub-Section (recursive key)

Category Entity:

- CategoryID
- CategoryName

Location Entity:

- Location ID
- Latitude

- Longitude
- Description

Project Entity:

- Project code (Primary key)
- Project name
- Project Description
- Date Started
- Date Completed

Staff Entity:

- Employee ID (Primary key)
- First name
- last name
- Date first Employed
- D.O.B
- Gender
- Postal Address
- Contact phone number
- Email Address
- Role name (Foreign key)

Role Entity:

- Role name (Primary key)
- Role description

Contracts Entity:

- Contract number (Primary key)
- Contract Name
- Contract Description
- Estimated cost
- Actual cost
- Date Started
- Date Ended

Contractor Entity:

- Company Name (Primary key)
- Company Address
- Contact details

Road Entity

(old)

```
CREATE TABLE Road (  
road_id NUMBER(6) PRIMARY KEY,  
road_name VARCHAR2(50) NOT NULL,  
road_desc VARCHAR2(200),  
road_length NUMBER(10) NOT NULL,  
sub_section VARCHAR(50),  
CONSTRAINT road_road_id_pk PRIMARY KEY (road_id),  
CONSTRAINT road_road_id_pk FOREIGN KEY (sub_section) REFERENCES road(road_id)  
);
```

(old_v2)

```
CREATE TABLE Road (  
road_id NUMBER(6),  
road_name VARCHAR2(50) NOT NULL,  
road_desc VARCHAR2(200),  
road_length NUMBER(10) NOT NULL,  
sub_section NUMBER(6),  
CONSTRAINT road_road_id_pk PRIMARY KEY (road_id),  
CONSTRAINT road_sub_section_fk FOREIGN KEY (sub_section) REFERENCES  
Road(road_id)  
);
```

(New)

```
CREATE TABLE Road (  
road_id VARCHAR2(6),  
road_name VARCHAR2(50) NOT NULL,  
road_desc VARCHAR2(200),  
road_length NUMBER(10) NOT NULL,  
sub_section VARCHAR2(6),  
CONSTRAINT road_road_id_pk PRIMARY KEY (road_id),  
CONSTRAINT road_sub_section_fk FOREIGN KEY (sub_section) REFERENCES  
Road(road_id)  
);
```

,

Name	Null?	Type
-----	-----	-----
ROAD_ID	NOT NULL	NUMBER(6)
ROAD_NAME	NOT NULL	VARCHAR2(50)
ROAD_DESC		VARCHAR2(200)
ROAD_LENGTH	NOT NULL	NUMBER(10)
SUB_SECTION		NUMBER(6)

Sample Road Entity:

Category of road type : main highway, secondary road, unsealed road

Road ID	Road Name	Road Description	Road Length	Sub-section
R-4080	Bucklands Beach	Asphalt	10	
R-4081	Buck Beach	Asphalt	5	R-4080
R-4082	Sunderland Beach	Asphalt	5	R-4080
R-1156	Elstree Avenue		12	
R-3560	State Highway 20	Tar	20	
R-3561	South-Western 12	Tar	12	R-3560
R-8150	Great South		5	
R-5910	Panama Road	Asphalt	10	
R-5911	McLennan Road	Asphalt	4	R-5910
R-3450	Te Kehu Way		10	

Category Entity

(old)

CREATE TABLE Category (
category_id NUMBER(6) PRIMARY KEY,

```
category_name VARCHAR2(30) NOT NULL  
);
```

(new)

```
CREATE TABLE category (  
category_id NUMBER(6),  
category_name VARCHAR2(30) NOT NULL,  
CONSTRAINT category_category_id_pk PRIMARY KEY (category_id)  
);
```

Done

Name	Null?	Type
CATEGORY_ID	NOT NULL	NUMBER(6)
CATEGORY_NAME	NOT NULL	VARCHAR2(30)

Sample Category Entity:

Category of road type : main highway, secondary road, unsealed road

Category ID	Category Name
01	Main Highway
02	Road
03	Secondary road
04	Unsealed Road
05	Rural road

Location Entity

```
CREATE TABLE location (  
location_id VARCHAR2(6) ,  
latitude NUMBER(4) NOT NULL,  
longitude NUMBER(4) NOT NULL,  
location_desc VARCHAR2(200),
```

```
CONSTRAINT location_location_id_pk PRIMARY KEY (location_id)
);
```

Name	Null?	Type
-----	-----	-----
LOCATION_ID	NOT NULL	VARCHAR2(6)
LATITUDE	NOT NULL	NUMBER(4)
LONGITUDE	NOT NULL	NUMBER(4)
LOCATION_DESC		VARCHAR2(200)

Sample Location Entity:

Location ID	Latitude	Longitude	Location Description
LOC-40	-36	174	Buckland
LOC-41	-36	156	Buckland
LOC-42	-36	140	Buckland
LOC-15	-12	70	Flatbush
LOC-30	40	-179	Mount Wellington
LOC-31	40	-154	Mount Wellington
LOC-80	-23	-123	Flatbush
LOC-10	-244	12	Albany
LOC-11	-244	12	Albany
LOC-34	-122	242	Penrose

Project Entity

```
CREATE TABLE project (
project_code VARCHAR2(6),
project_name VARCHAR(40) NOT NULL,
project_desc VARCHAR(200),
```



```

date_started DATE NOT NULL,
date_completed DATE,
CONSTRAINT project_project_code_pk PRIMARY KEY (project_code)
);

```

```

<new code>
CREATE TABLE project (
project_code VARCHAR2(6),
project_name VARCHAR(40) NOT NULL,
project_desc VARCHAR(200),
date_started DATE NOT NULL,
date_completed DATE,
contract_num VARCHAR2(7),
CONSTRAINT project_project_code_pk PRIMARY KEY (project_code),
CONSTRAINT project_contract_num_fk FOREIGN KEY (contract_num)
REFERENCES contract(contract_num)
);

```

Sample Project Entity:

Project Code	Project Name	Project Description	Date started	Date completed	Contract_num
F4KJ3Q	Melted Butter	Repaving road	20/10/24		C-90967
NVJWQR	Smooth rock	Repaving road	10/09/24	1/10/24	C-12356
P4ETHJ	Evergreen	Road ExtensionNV	8/11/24	20/01/25	C-86064
4ZCG7Z	Project Gunn	Repaving road	20/10/24	26/10/24	C-12563
GKFVAQ	Grounds	Road Repairs	11/09/24	22/10/24	C-78643
TYQ8AQ	Tree	Road Extension	12/09/24	28/10/24	C-56789
AB7A3M	Waterflow		20/10/24		C-12134

Name	Null?	Type
PROJECT_CODE	NOT NULL	VARCHAR2 (6)
PROJECT_NAME	NOT NULL	VARCHAR2 (40)
PROJECT_DESC		VARCHAR2 (200)
DATE_STARTED	NOT NULL	DATE
DATE_COMPLETED		DATE
CONTRACT_NUM		VARCHAR2 (7)

Staff Entity

(old)

```
CREATE TABLE staff (
Employee_id VARCHAR2(5) PRIMARY KEY,
Firstname VARCHAR2(20) NOT NULL,
Lastname VARCHAR2(20) NOT NULL,
Employment_date DATE NOT NULL,
DOB DATE NOT NULL,
Gender CHAR(1),
street VARCHAR2(40) NOT NULL,
city VARCHAR2(25) NOT NULL,
zipcode NUMBER(4) NOT NULL,
Phonenum NUMBER(10) NOT NULL,
Email VARCHAR2(20)
);
```

(new)

```
CREATE TABLE staff (
employee_id VARCHAR2(6),
firstname VARCHAR2(20) NOT NULL,
lastname VARCHAR2(20) NOT NULL,
employment_date DATE NOT NULL,
dob DATE NOT NULL,
gender CHAR(1),
street VARCHAR2(40) NOT NULL,
city VARCHAR2(25) NOT NULL,
zipcode NUMBER(4) NOT NULL,
phone_num NUMBER(10) NOT NULL,
```

email VARCHAR2(20);
CONSTRAINT staff_employee_id_pk PRIMARY KEY (employee_id)
;

Name	Null?	Type
EMPLOYEE_ID	NOT NULL	VARCHAR2(5)
FIRSTNAME	NOT NULL	VARCHAR2(20)
LASTNAME	NOT NULL	VARCHAR2(20)
EMPLOYMENT_DATE	NOT NULL	DATE
DOB	NOT NULL	DATE
GENDER		CHAR(1)
STREET	NOT NULL	VARCHAR2(40)
CITY	NOT NULL	VARCHAR2(25)
ZIPCODE	NOT NULL	NUMBER(4)
PHONENUM	NOT NULL	NUMBER(10)
EMAIL		VARCHAR2(20)

Sample Staff Entity:

Employee ID	First Name	Last name	Employment Date	Date of Birth	Gender	Street	city	Zipcode	Phone Number	Email
EP-001	Bob	Forest	20/09/22	02/01/00	M	224B Clare Street	Auckland	1234	0924797635	BOBFORST
EP-002	Welis	Arrow	20/09/22	06/02/89	M	643B Crane Road	Auckland	5675	0273748284	WEAROW
EP-101	Carl	Waters	12/01/23	16/08/95	M	45C Steward rd	Auckland	3456	0776439439	CAWATERS
EP-201	Jim	Karl	15/09/23	12/05/95		89A Bram Street	Auckland	8459	0355772984	JIKARL
EP-202	Ale	Birch	20/09/23	27/03/97	M	12N Quartz Street	Auckland	4253	0822935233	ALEBIRCH
EP-203	Toast	Birch	20/09/23	27/03/97	F	12N Quartz Street	Auckland	4253	0988569987	TOABIRCH
EP-021	Beth	Sun	30/11/22	19/11/01	F	56 Steele Road	Auckland	9074	0766546887	BESUN
EP-221	Crane	Gunn	4/11/23	20/12/94	M	37Z Cram Street	Auckland	3200	0234567876	CRGUNN
EP-081	York	Joseph	15/12/22	21/01/87	M	121 Rata Road	Auckland	5019	0123456782	YOJOSEPH
EP-012	Arnold	Swan	26/11/22	13/04/90		78Y Lorden Road	Auckland	2234	0647548476	ARSWAN
EP-110	Roger	Mary	12/08/22	16/08/78		190 Monarch Avenue	Auckland	7809	0912649154	ROMARY
EP-025	Mike	Hunt	10/06/22	19/02/99	M	123 Abcde Road	Auckland	7654	0234567853	MIHUNT
EP-134	Migel	Santos	23/07/23	29/09/89		29 Weir Grove	Auckland	9812	0981490186	MISANTOS

EP-122	Anderi	Zeph	12/05/22	12/06/00	M	15 Swynford Lane	Auckland	1827	0765926541	ANZEPH
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Role Entity

```
CREATE TABLE role (  
  role_name VARCHAR2(20) NOT NULL,  
  role_desc VARCHAR2(200),  
  CONSTRAINT role_role_name_pk PRIMARY KEY (role_name)  
);
```

Sample Role Entity:

Role Name	Role Description
Manager	Accepts contracts and manages all Road worker roles
Road Leader	Manages Road Workers
Road Machine Operator	Uses machinery to work on road
Road Traffic Controller	Redirect traffic away from roadworks
Road Worker	Assists in Road works

```
Name      Null?   Type  
-----  
ROLE_NAME NOT NULL VARCHAR2(20)  
ROLE_DESC          VARCHAR2(200)
```

Contracts Entity

```
(old)  
CREATE TABLE contract (  
  contract_num NUMBER(10),  
  contract_desc VARCHAR2(200),  
  estimated_cost NUMBER(20) NOT NULL,
```

```
actual_cost NUMBER(20) NOT NULL,  
date_started DATE NOT NULL,  
date_ended DATE,  
company_name VARCHAR2(15),  
role_name VARCHAR2(20),  
CONSTRAINT contract_contract_num_pk PRIMARY KEY (contract_num),  
CONSTRAINT contract_role_name_fk FOREIGN KEY (role_name) REFERENCES  
role(role_name),  
CONSTRAINT contract_company_name_fk FOREIGN KEY (company_name) REFERENCES  
contractor(company_name)  
);
```

(old)

```
CREATE TABLE contract (  
contract_num VARCHAR2(7) NOT NULL,  
contract_desc VARCHAR2(200),  
estimated_cost NUMBER(20) NOT NULL,  
actual_cost NUMBER(20) NOT NULL,  
date_started DATE NOT NULL,  
date_ended DATE,  
company_name VARCHAR2(15) NOT NULL,
```

```
CONSTRAINT contract_contract_num_pk PRIMARY KEY (contract_num),  
CONSTRAINT contract_company_name_fk FOREIGN KEY (company_name) REFERENCES  
contractor(company_name)  
);
```

(new)

```
CREATE TABLE contract (  
contract_num VARCHAR2(7) NOT NULL,  
contract_name VARCHAR2(50),  
contract_desc VARCHAR2(200),  
estimated_cost NUMBER(20) NOT NULL,  
actual_cost NUMBER(20) NOT NULL,  
date_started DATE NOT NULL,  
date_ended DATE,  
company_name VARCHAR2(15) NOT NULL,  
CONSTRAINT contract_contract_num_pk PRIMARY KEY (contract_num),  
CONSTRAINT contract_company_name_fk FOREIGN KEY (company_name) REFERENCES  
contractor(company_name)  
);
```

Sample Contracts Entity:

Contract Num	Contract Name	Contract Description	Estimated Cost	Actual Cost	Date Started	Date Ended	Company Name
C-90967	Ford Contract	Medium-term	120000	90000	20/10/24		Ford Harrison LTD
C-12356	Buggy Contract Agreement	Short-term	36000	42000	10/09/24	1/10/24	Buggy Machinery Roder
C-86064	Roder Contract	Long-term	450000	200000	8/11/24	20/01/25	Roder Fixer LTD
C-12563	Harvey Road Repair agreement	Long-term	786000	740000	20/10/24	26/10/24	Harvey Construction
C-78643	Noel Road Contract		20000	40000	11/09/24	22/10/24	Noel Road LTD
C-56789	Ford Contract 2	Short-term	12000	20000	12/09/24	28/10/24	Ford Harrison LTD
C-12134	Noel Road Contract Agreement		59000	59000	20/10/24		Noel Road LTD

Name	Null?	Type
CONTRACT_NUM	NOT NULL	VARCHAR2(7)
CONTRACT_DESC		VARCHAR2(200)
ESTIMATED_COST	NOT NULL	NUMBER(20)
ACTUAL_COST	NOT NULL	NUMBER(20)
DATE_STARTED	NOT NULL	DATE
DATE_ENDED		DATE
COMPANYNAME	NOT NULL	VARCHAR2(15)

Contractor Entity

```
CREATE TABLE contractor (  
  company_name VARCHAR2(40) NOT NULL,  
  company_address VARCHAR2(50) NOT NULL,  
  contactdetails NUMBER(10) NOT NULL  
  
);  
  
(new)  
CREATE TABLE contractor (
```

```

company_name VARCHAR2(40),
company_address VARCHAR2(50) NOT NULL,
contactdetails NUMBER(10) NOT NULL,
CONSTRAINT contractor_company_name_pk PRIMARY KEY (company_name)
);

```

Sample Contractor Entity:

Company Name	Company Address	Contact Details
Ford Harrison LTD	Saint Georges Road	0788956431
Roader Fixer LTD	55 Wellesley Street East	0129876409
Harvey Construction	67 Crambold Street	0567894223
Noel Road LTD	Ham Ridge Road	0226578943
Buggy Machinery Roader	90A Quarry St East	0227654386

```

Name          Null?   Type
-----
COMPANY_NAME  NOT NULL VARCHAR2(15)
COMPANY_ADDRESS NOT NULL VARCHAR2(50)
CONTACTDETAILS NOT NULL NUMBER(10)

```

RoadCategory (Bridging Tables)

(old)

```

CREATE TABLE RoadCategory (
roadsroadid CHAR(10) NOT NULL,
categorycategoryid (10) NOT NULL,

```

```

CONSTRAINT roadcategory_roadsroadid_fk FOREIGN KEY (roadsroadid) REFERENCES
roads(roadid),

```

```

CONSTRAINT roadcategory_categorycategoryid_fk FOREIGN KEY
(categorycategoryid) REFERENCES category(categoryid)
);

```

(new)

```

CREATE TABLE roadcategory (
road_id VARCHAR2(6),

```

```

category_id NUMBER(6),
CONSTRAINT roadcategory_road_id_category_id_pk PRIMARY KEY (road_id, category_id),
CONSTRAINT roadcategory_road_id_fk FOREIGN KEY (road_id) REFERENCES
road(road_id),
CONSTRAINT roadcategory_category_id_fk FOREIGN KEY(category_id)
REFERENCES category(category_id)
);

```

Sample Road Category:

Road ID	Category ID
R-4080	02
R-4081	02
R-4082	02
R-1156	04
R-3560	01
R-3561	01
R-8150	02
R-5910	03
R-5911	03
R-3450	05

StartEnd (Bridging Tables)

(old)

```

CREATE TABLE StartEnd (
locationstart VARCHAR(20) NOT NULL,
locationend VARCHAR(20) NOT NULL,
locationlocationid NUMBER(6) NOT NULL,
roadsroadid NUMBER(6) NOT NULL,
CONSTRAINT startend_locationlocationid_fk FOREIGN KEY (locationlocationid)
REFERENCES location(locationid),

```



```
CONSTRAINT startend_roadsroadid_fk FOREIGN KEY
(roadsroadid) REFERENCES roads(roadid)
);
```

```
(new)
CREATE TABLE startend (
location_id VARCHAR2(6),
road_id VARCHAR2(6),
locationstart VARCHAR(20) NOT NULL,
locationend VARCHAR(20) NOT NULL,
CONSTRAINT startend_location_id_road_id_pk PRIMARY KEY (location_id, road_id),
CONSTRAINT startend_location_id_fk FOREIGN KEY (location_id) REFERENCES
location(location_id),
CONSTRAINT startend_road_id_fk FOREIGN KEY(road_id) REFERENCES road(road_id)
);
```

StartEnd (Bridging Tables)

Road ID	Location ID	Location Start	Location End
R-4080	LOC-40	1 Bucklands Beach	35 Bucklands Beach
R-4081	LOC-41	1 Buck Beach	44 Buck Beach
R-4082	LOC-42	1 Sunderland Beach	50 Sunderland Beach
R-1156	LOC-15	1 Elstree Avenue	20 Elstree Avenue
R-3560	LOC-30	State Highway 20	State Highway 20B
R-3561	LOC-31	South-Western 12	South-Western 12B
R-8150	LOC-80	1 Great South	101 Great South
R-5910	LOC-10	1 Panama Road	25 Panama Road
R-5911	LOC-11	1 McLennan Road	40 McLennan Road
R-3450	LOC-34	1 Te Kehu Way	24 Te Kehu Way

Record (Bridging Tables)

(old)

```
CREATE TABLE Record (
projectproject_code VARCHAR(10) NOT NULL,
roadsroadid VARCHAR(10) NOT NULL,
CONSTRAINT record_projectproject_code_fk FOREIGN KEY (projectproject_code)
REFERENCES project(project_code),
CONSTRAINT record_roadsroadid_fk FOREIGN KEY
(roadsroadid) REFERENCES roads(roadid)
);
```

```
(new)
CREATE TABLE record (
project_code VARCHAR2(6),
road_id VARCHAR2(6),
CONSTRAINT record_project_code_road_id_pk PRIMARY KEY (project_code, road_id),
CONSTRAINT record_project_code_fk FOREIGN KEY (project_code) REFERENCES
project(project_code),
CONSTRAINT record_road_id_fk FOREIGN KEY(road_id) REFERENCES road(road_id)
);
```

Record (Bridging Tables) Sampling

Project Code	Road ID
F4KJ3Q	R-4080
F4KJ3Q	R-4081
NVJWQR	R-4082
P4ETHJ	R-1156
4ZCG7Z	R-3560
4ZCG7Z	R-3561
GKFVAQ	R-8150
TYQ8AQ	R-5910
TYQ8AQ	R-5911
AB7A3M	R-3450

Staff_record (Bridging Tables)

(old)

```
CREATE TABLE Staff_record (  
  projectproject_code VARCHAR(10) NOT NULL,  
  staffemployeeid VARCHAR(10) NOT NULL,  
  CONSTRAINT staff_record_projectproject_code_fk FOREIGN KEY (projectproject_code)  
  REFERENCES project(project_code),  
  CONSTRAINT staff_record_staffemployeeid_fk FOREIGN KEY  
  (staffemployeeid) REFERENCES staff(employeeid)  
);
```

(new)

```
CREATE TABLE staff_record (  
  project_code VARCHAR2(6),  
  employee_id VARCHAR2(6),  
  CONSTRAINT staff_record_project_code_employee_id PRIMARY KEY (project_code,  
  employee_id),  
  CONSTRAINT staff_record_project_code_fk FOREIGN KEY (project_code) REFERENCES  
  project(project_code),  
  CONSTRAINT staff_record_staff_employee_id_fk FOREIGN KEY (employee_id)  
  REFERENCES staff(employee_id)  
);
```

Sample Staff_record:

Project Code	Employee ID
F4KJ3Q	EP-001
F4KJ3Q	EP-002
F4KJ3Q	EP-101
NVJWQR	EP-201
NVJWQR	EP-202
4ZCG7Z	EP-203
P4ETHJ	EP-021
GKFVAQ	EP-221
P4ETHJ	EP-081

4ZCG7Z	EP-012
GKFVAQ	EP-110
TYQ8AQ	EP-025
AB7A3M	EP-134
AB7A3M	EP-122

StaffRole (Bridging Tables)

(old)

```
CREATE TABLE StaffRole (
  rolerolename VARCHAR(30) NOT NULL,
  staffemployeeid VARCHAR(10) NOT NULL,
  assigned_date DATE,
  assigned_enddate DATE,
  CONSTRAINT staffrole_rolerolename_fk FOREIGN KEY (rolerolename) REFERENCES
  role(rolename),
  CONSTRAINT staffrole_staffemployeeid_fk FOREIGN KEY
  (staffemployeeid) REFERENCES staff(employeeid)
);
```

(new)

```
CREATE TABLE staffrole (
  role_name VARCHAR2(20),
  employee_id VARCHAR2(6),
  assigned_date DATE NOT NULL,
  assigned_enddate DATE,
  CONSTRAINT staff_role_role_name_employee_id PRIMARY KEY (role_name, employee_id),
  CONSTRAINT staffrole_role_name_fk FOREIGN KEY (role_name) REFERENCES
  role(role_name),
  CONSTRAINT staffrole_employee_id_fk FOREIGN KEY (employee_id) REFERENCES
  staff(employee_id)
);
```

StaffRole Sampling

Role Name	Employee ID	Assigned Date	Assigned End Date
-----------	-------------	---------------	-------------------

Road Leader	EP-001	20/09/22	16/08/23
Manager	EP-001	16/08/23	
Road Worker	EP-002	20/09/22	14/07/23
Road Machine Operator	EP-002	14/07/23	
Road Machine Operator	EP-101	12/01/23	
Road Worker	EP-201	15/09/23	20/12/23
Road Machine Operator	EP-201	20/12/23	
Road Worker	EP-202	20/09/23	
Road Worker	EP-203	20/09/23	
Manager	EP-021	31/11/22	16/08/23
Road Leader	EP-021	16/08/23	
Road Worker	EP-221	4/11/23	
Road Worker	EP-081	15/12/22	
Road Worker	EP-012	26/11/22	
Road Traffic Controller	EP-110	12/08/22	
Road Traffic Controller	EP-025	10/06/22	
Road Traffic Controller	EP-134	23/07/23	
Road Traffic Controller	EP-122	12/05/22	

Entry Roads

SELECT * FROM road;

```
INSERT INTO ROAD (road_id, road_name, road_desc, road_length, sub_section)
VALUES ('&road_id', '&road_name', '&road_desc', '&road_length', '&sub_section');
```

	ROAD_ID	ROAD_NAME	ROAD_DESC	ROAD_LENGTH	SUB_SECTION
1	R-4080	Bucklands Beach	Asphalt	10	(null)
2	R-4081	Buck Beach	Asphalt	5	R-4080
3	R-4082	Sunderland Beach	Asphalt	5	R-4080
4	R-1156	Elstree Avenue	(null)	12	(null)
5	R-3560	State Highway 20	Tar	20	(null)
6	R-3561	South-Western 12	Tar	12	R-3560
7	R-8150	Great South	(null)	5	(null)
8	R-5910	Panama Road	Asphalt	10	(null)
9	R-5911	McLennan Road	Asphalt	4	R-5910
10	R-3450	Te Kehu Way	(null)	10	(null)

```
INSERT INTO ROAD
(road_id, road_name, road_desc, road_length, sub_section)
VALUES ('R-4080', 'Bucklands Beach', 'Asphalt', 10, NULL);
```

```
INSERT INTO ROAD
(road_id, road_name, road_desc, road_length, sub_section)
VALUES ('R-4081', 'Buck Beach', 'Asphalt', 5, 'R-4080');
```

```
INSERT INTO ROAD
(road_id, road_name, road_desc, road_length, sub_section)
VALUES ('R-4082', 'Sunderland Beach', 'Asphalt', 5, 'R-4080');
```

```
INSERT INTO ROAD
(road_id, road_name, road_desc, road_length, sub_section)
VALUES ('R-1156', 'Elstree Avenue', NULL, 12, NULL);
```

```
INSERT INTO ROAD
(road_id, road_name, road_desc, road_length, sub_section)
VALUES ('R-3560', 'State Highway 20', 'Tar', 20, NULL);
```

```
INSERT INTO ROAD
(road_id, road_name, road_desc, road_length, sub_section)
VALUES ('R-3561', 'South-Western 12', 'Tar', 12, 'R-3560');
```

```
INSERT INTO ROAD
(road_id, road_name, road_desc, road_length, sub_section)
VALUES ('R-8150', 'Great South', NULL, 5, NULL);
```

```
INSERT INTO ROAD
(road_id, road_name, road_desc, road_length, sub_section)
VALUES ('R-5910', 'Panama Road', 'Asphalt', 10, NULL);
```

```
INSERT INTO ROAD
(road_id, road_name, road_desc, road_length, sub_section)
VALUES ('R-5911', 'McLennan Road', 'Asphalt', 4, 'R-5910');
```

```
INSERT INTO ROAD
(road_id, road_name, road_desc, road_length, sub_section)
VALUES ('R-3450', 'Te Kehu Way', NULL, 10, NULL);
```

```
COMMIT;
```

Entry Category

```
SELECT * FROM category;
INSERT INTO category (category_id, category_name)
VALUES ('&category_id', '&category_name');
COMMIT;
```

	CATEGORY_ID	CATEGORY_NAME
1	1	Main Highway
2	2	Road
3	3	Secondary road
4	4	Unsealed road
5	5	Rural road

```
INSERT INTO CATEGORY
(category_id, category_name)
VALUES
(01, 'Main Highway');
```

```
INSERT INTO CATEGORY
(category_id, category_name)
VALUES
```

```
(02, 'Road');
```

```
INSERT INTO CATEGORY  
(category_id, category_name)  
VALUES  
(03, 'Secondary Road');
```

```
INSERT INTO CATEGORY  
(category_id, category_name)  
VALUES  
(04, 'Unsealed Road');
```

```
INSERT INTO CATEGORY  
(category_id, category_name)  
VALUES  
(05, 'Rural Road');  
COMMIT;
```

```
—
```

Entry Location

```
INSERT INTO LOCATION  
(location_id, latitude, longitude, location_desc)  
VALUES  
(LOC-40, -36, 174, 'Buckland');
```

```
INSERT INTO LOCATION  
(location_id, latitude, longitude, location_desc)  
VALUES  
(LOC-41, -36, 156, 'Buckland');
```

```
INSERT INTO LOCATION  
(location_id, latitude, longitude, location_desc)  
VALUES  
(LOC-42, -36, 140, 'Buckland');
```

```
INSERT INTO LOCATION  
(location_id, latitude, longitude, location_desc)  
VALUES  
(LOC-15, -12, 70, 'Flatbush');
```



```
INSERT INTO LOCATION
(location_id, latitude, longitude, location_desc)
VALUES
('LOC-30', 40, -179, 'Mount Wellington');
```

```
INSERT INTO LOCATION
(location_id, latitude, longitude, location_desc)
VALUES
('LOC-31', 40, -154, 'Mount Wellington');
```

```
INSERT INTO LOCATION
(location_id, latitude, longitude, location_desc)
VALUES
('LOC-80', -23, -123, 'Flatbush');
```

```
INSERT INTO LOCATION
(location_id, latitude, longitude, location_desc)
VALUES
('LOC-10', -244, 12, 'Albany');
```

```
INSERT INTO LOCATION
(location_id, latitude, longitude, location_desc)
VALUES
('LOC-11', -244, 12, 'Albany');
```

```
INSERT INTO LOCATION
(location_id, latitude, longitude, location_desc)
VALUES
('LOC-34', -122, 242, 'Penrose');
COMMIT;
```

	LOCATION_ID	LATITUDE	LONGITUDE	LOCATION_DESC
1	LOC-80	-23	-123	FLATBUSH
2	LOC-40	-36	174	Buckland
3	LOC-41	-36	156	Buckland
4	LOC-42	-36	140	Buckland
5	LOC-15	-12	70	Flatbush
6	LOC-30	40	-179	Mount Wellington
7	LOC-31	40	-154	Mount Wellington
8	LOC-10	-244	12	Albany
9	LOC-11	-244	12	Albany
10	LOC-34	-122	242	Penrose

Entry Project

(old)

```
INSERT INTO PROJECT (project_code, project_name, project_desc,
date_started, date_completed)
VALUES ('&project_code', '&project_name', '&project_desc', '&date_started',
'&date_completed');
```

(new)

```
INSERT INTO PROJECT (project_code, project_name, project_desc,
date_started, date_completed, contract_num)
VALUES ('&project_code', '&project_name', '&project_desc', '&date_started',
'&date_completed', '&contract_num');
```

(old)

```

5 SELECT * FROM PROJECT;
5 INSERT INTO PROJECT (project_code, project_name, project_desc,
7 date_started, date_completed)
3 VALUES ('&project_code', '&project_name', '&project_desc', '&date_started',
3 '&date_completed');
0

```

Script Output x | Autotrace x | Query Result x

SQL | All Rows Fetched: 7 in 0.011 seconds

	PROJECT_CODE	PROJECT_NAME	PROJECT_DESC	DATE_STARTED	DATE_COMPLETED
1	F4KJ3Q	Melted Butter	Repaving road	20/10/24	(null)
2	NVJWQR	Smooth rock	Repaving road	10/09/24	01/10/24
3	P4ETHJ	Evergreen	Road ExtensionNV	08/11/24	20/01/25
4	4ZCG7Z	Project Gunn	Repaving road	20/10/24	26/10/24
5	GKFVAQ	Grounds	Road Repairs	11/09/24	22/10/24
6	TYQ8AQ	Tree	Road Extension	12/09/24	28/10/24
7	AB7A3M	Waterflow	(null)	20/10/24	(null)

(new)

```

72 SELECT * FROM PROJECT;
73 INSERT INTO PROJECT (project_code, project_name, project_desc,
74 date_started, date_completed)
75 VALUES ('&project_code', '&project_name', '&project_desc', '&date_started',
76 '&date_completed');
77 Describe project;
78
79
80 --Staff

```

Script Output x | Autotrace x | Query Result x | Query Result 1 x

SQL | All Rows Fetched: 7 in 0.012 seconds

	PROJECT_CODE	PROJECT_NAME	PROJECT_DESC	DATE_STARTED	DATE_COMPLETED	CONTRACT_NUM
1	F4KJ3Q	Melted Butter	Repaving road	20/10/24	(null)	C-90967
2	NVJWQR	Smooth rock	Repaving road	10/09/24	01/10/24	C-12356
3	P4ETHJ	Evergreen	Road ExtensionNV	08/11/24	20/01/25	C-86064
4	4ZCG7Z	Project Gunn	Repaving road	20/10/24	26/10/24	C-12563
5	GKFVAQ	Grounds	Road Repairs	11/09/24	22/10/24	C-78643
6	TYQ8AQ	Tree	Road Extension	12/09/24	28/10/24	C-56789
7	AB7A3M	Waterflow	(null)	20/10/24	(null)	C-12134

INSERT INTO PROJECT

(project_code, project_name, project_desc, date_started, date_completed)

```

VALUES
('F4KJ3Q', 'Melted Butter', 'Repaving Road', '20-10-24', NULL);

(project_code, project_name, project_desc, date_started, date_completed)
VALUES
('NVJWQR', 'Smooth rock', 'Repaving Road', '10-09-24', '1-10-24');

(project_code, project_name, project_desc, date_started, date_completed)
VALUES
('P4ETHJ', 'Evergreen', 'Road Extension', '8-11-24', '20-01-25');

(project_code, project_name, project_desc, date_started, date_completed)
VALUES
('4ZCG7Z', 'Project Gunn', 'Repaving Road', '20-10-24', '26-10-24');

(project_code, project_name, project_desc, date_started, date_completed)
VALUES
('GKFVAQ', 'Grounds', 'Road Repairs', '11-09-24', '22-10-24');

(project_code, project_name, project_desc, date_started, date_completed)
VALUES
('TYQ8AQ', 'Tree', 'Road Extension', '12-09-24', '28-10-24');

(project_code, project_name, project_desc, date_started, date_completed)
VALUES
('AB7A3M', 'Waterflow', NULL, '20-10-24', NULL);

COMMIT;

```

Entry Staff

```

CREATE TABLE staff (
employee_id VARCHAR2(6),
firstname VARCHAR2(20) NOT NULL,
lastname VARCHAR2(20) NOT NULL,
employment_date DATE NOT NULL,
dob DATE NOT NULL,
gender CHAR(1),
street VARCHAR2(40) NOT NULL,
city VARCHAR2(25) NOT NULL,
zipcode NUMBER(4) NOT NULL,
phone_num NUMBER(10) NOT NULL,
email VARCHAR2(20),

```

```
CONSTRAINT staff_employee_id_pk PRIMARY KEY (employee_id)
);
drop table staff;
desc staff;
```

```
INSERT INTO STAFF
(employee_id, firstname, lastname, employment_date, dob, gender, street, city, zipcode,
phone_num, email)
VALUES
('EP-001', 'Bob', 'Forest', '20-09-22', '02-01-00', 'M', '224B Clare Street', 'Auckland', 1234,
0924797635, 'BOBFOREST');
```

```
INSERT INTO STAFF
(employee_id, firstname, lastname, employment_date, dob, gender, street, city, zipcode,
phone_num, email)
VALUES
('EP-002', 'Welis', 'Arow', '20-09-22', '06-02-89', 'M', '643B Crane Road', 'Auckland', 5675,
0273748284, 'WEAROW');
```

```
INSERT INTO STAFF
(employee_id, firstname, lastname, employment_date, dob, gender, street, city, zipcode,
phone_num, email)
VALUES
('EP-101', 'Carl', 'Waters', '12-01-23', '16-08-95', 'M', '45C Steward Rd', 'Auckland', 3456,
0776439439, 'CAWATERS');
```

```
INSERT INTO STAFF
(employee_id, firstname, lastname, employment_date, dob, gender, street, city, zipcode,
phone_num, email)
VALUES
('EP-201', 'Jim', 'Karl', '15-09-23', '12-05-95', null, '89A Bram Street', 'Auckland', 8459,
0355772984, 'JIKARL');
```

```
INSERT INTO STAFF
(employee_id, firstname, lastname, employment_date, dob, gender, street, city, zipcode,
phone_num, email)
VALUES
('EP-202', 'Ale', 'Birch', '20-09-23', '27-03-97', 'M', '12N Quartz Street', 'Auckland', 4253,
0822935233, 'ALEBIRCH');
```

```
INSERT INTO STAFF
(employee_id, firstname, lastname, employment_date, dob, gender, street, city, zipcode,
phone_num, email)
VALUES
```

```
('EP-203', 'Toast', 'Birch', '20-09-23', '27-03-97', 'F', '12N Quartz Street', 'Auckland', 4253, 0988569987, 'TOABIRCH');
```

```
INSERT INTO STAFF
```

```
(employee_id, firstname, lastname, employment_date, dob, gender, street, city, zipcode, phone_num, email)
```

```
VALUES
```

```
('EP-021', 'Beth', 'Sun', '30-11-22', '19-11-01', 'F', '56 Steele Road', 'Auckland', 9074, 0766546887, 'BESUN');
```

```
INSERT INTO STAFF
```

```
(employee_id, firstname, lastname, employment_date, dob, gender, street, city, zipcode, phone_num, email)
```

```
VALUES
```

```
('EP-221', 'Crane', 'Gunn', '4-11-23', '20-12-94', 'M', '37Z Cram Street', 'Auckland', 3200, 0234567876, 'CRGUNN');
```

```
INSERT INTO STAFF
```

```
(employee_id, firstname, lastname, employment_date, dob, gender, street, city, zipcode, phone_num, email)
```

```
VALUES
```

```
('EP-081', 'York', 'Joseph', '15-12-22', '21-01-87', 'M', '121 Rata Road', 'Auckland', 5019, 0123456782, 'YOJOSEPH');
```

```
INSERT INTO STAFF
```

```
(employee_id, firstname, lastname, employment_date, dob, gender, street, city, zipcode, phone_num, email)
```

```
VALUES
```

```
('EP-012', 'Arnold', 'Swan', '26-11-22', '13-04-90', NULL, '78Y Lorden Road', 'Auckland', 2234, 0647548476, 'ARSWAN');
```

```
INSERT INTO STAFF
```

```
(employee_id, firstname, lastname, employment_date, dob, gender, street, city, zipcode, phone_num, email)
```

```
VALUES
```

```
('EP-110', 'Roger', 'Mary', '12-08-22', '16-08-78', NULL, '190 Monarch Avenue', 'Auckland', 7809, 0912649154, 'ROMARY');
```

```
INSERT INTO STAFF
```

```
(employee_id, firstname, lastname, employment_date, dob, gender, street, city, zipcode, phone_num, email)
```

```
VALUES
```

```
('EP-025', 'Mike', 'Hunt', '10-06-22', '19-02-99', 'M', '123 Abcde Road', 'Auckland', 7654, 0234567853, 'MIHUNT');
```

INSERT INTO STAFF

(employee_id, firstname, lastname, employment_date, dob, gender, street, city, zipcode, phone_num, email)

VALUES

('EP-134', 'Migel', 'Santos', '23-07-23', '29-09-89', NULL, '29 Weir Grove', 'Auckland', 9812, 0981490186, 'MISANTOS');

INSERT INTO STAFF

(employee_id, firstname, lastname, employment_date, dob, gender, street, city, zipcode, phone_num, email)

VALUES

('EP-122', 'Anderi', 'Zeph', '12-05-22', '12-06-00', 'M', '15 Swynford Lane', 'Auckland', 1827, 0765926541, 'ANZEPH');

COMMIT;

17 SELECT * FROM STAFF;

18

Script Output x Autotrace x Query Result x Query Result 1 x

SQL All Rows Fetched: 14 in 0.019 seconds

EMPLOYEE_ID	FIRSTNAME	LASTNAME	EMPLOYMENT_DATE	DOB	GENDER	STREET	CITY	ZIPCODE	PHONE_NUM	EMAIL
1 EP-001	Bob	Forest	20/09/22	02/01/00	M	224B Clare Street	Auckland	1234	924797635	BOBFORST
2 EP-002	Wellis	Arow	20/09/22	06/02/89	M	643B Crane Road	Auckland	5675	273748284	WEAROW
3 EP-101	Carl	Waters	12/01/23	16/08/95	M	45C Steward Rd	Auckland	3456	776439439	CAWATERS
4 EP-201	Jim	Karl	15/09/23	12/05/95	(null)	89A Bram Street	Auckland	8459	355772984	JKARL
5 EP-202	Ale	Birch	20/09/23	27/03/97	M	12N Quartz Street	Auckland	4253	822935233	ALEBIRCH
6 EP-203	Toast	Birch	20/09/23	27/03/97	F	12N Quartz Street	Auckland	4253	988569987	TOABIRCH
7 EP-021	Beth	Sun	30/11/22	19/11/01	F	56 Steele Road	Auckland	9074	766546887	BESUN
8 EP-221	Crane	Gunn	04/11/23	20/12/94	M	37Z Cram Street	Auckland	3200	234567876	CRGUNN
9 EP-081	York	Joseph	15/12/22	21/01/87	M	121 Rata Road	Auckland	5019	123456782	YOJOSEPH
10 EP-012	Arnold	Swan	26/11/22	13/04/90	(null)	78Y Lorden Road	Auckland	2234	647548476	ARSWAN
11 EP-110	Roger	Mary	12/08/22	16/08/78	(null)	190 Monarch Avenue	Auckland	7809	912649154	ROMARY
12 EP-025	Mike	Hunt	10/06/22	19/02/99	M	123 Abode Road	Auckland	7654	234567853	MIHUNT
13 EP-134	Migel	Santos	23/07/23	29/09/89	(null)	29 Weir Grove	Auckland	9812	981490186	MISANTOS
14 EP-122	Anderi	Zeph	12/05/22	12/06/00	M	15 Swynford Lane	Auckland	1827	765926541	ANZEPH

Entry Role

INSERT INTO ROLE (role_name, role_desc)

VALUES ('Manager', 'Accepts contracts and manages all Road worker roles');

INSERT INTO ROLE (role_name, role_desc)

VALUES ('Road Leader', 'Manages Road Workers');

INSERT INTO ROLE (role_name, role_desc)

VALUES ('Road Machine Operator', 'Uses machinery to work on road');

INSERT INTO ROLE (role_name, role_desc)

VALUES ('Road Traffic Controller', 'Redirect traffic away from roadworks');

INSERT INTO ROLE (role_name, role_desc)

VALUES ('Road Worker', 'Assists in Road work');

COMMIT;

	ROLE_NAME	ROLE_DESC
1	Manager	Accepts contracts and manages all Road worker roles
2	Road Leader	Manages Road Workers
3	Road Worker	Assists in Road work
4	Road Machine Operator	Uses machinery to work on road
5	Road Traffic Controller	Redirect traffic away from roadworks

Entry Contract

(old)

INSERT INTO CONTRACT

(contract_num, contract_desc, estimated_cost, actual_cost, date_started, date_ended,
company_name)

VALUES

('C-90967', 'Medium-term', 120000, 90000, '20-10-24', NULL, 'Ford Harrison LTD');

INSERT INTO CONTRACT

(contract_num, contract_desc, estimated_cost, actual_cost, date_started, date_ended,
company_name)

VALUES

('C-12356', 'Short-term', 36000, 42000, '10-09-24', '01-10-24', 'Buggy Machinery Roader');

INSERT INTO CONTRACT

(contract_num, contract_desc, estimated_cost, actual_cost, date_started, date_ended,
company_name)

VALUES

('C-86064', 'Long-term', 450000, 200000, '08-11-24', '20-01-25', 'Roader Fixer LTD');

INSERT INTO CONTRACT

(contract_num, contract_desc, estimated_cost, actual_cost, date_started, date_ended,
company_name)

VALUES

('C-12563', 'Long-term', 786000, 740000, '20-10-24', '26-10-24', 'Harvey Construction');

INSERT INTO CONTRACT

(contract_num, contract_desc, estimated_cost, actual_cost, date_started, date_ended,
company_name)

VALUES

('C-78643', NULL, 20000, 40000, '11-09-24', '22-10-24', 'Noel Road LTD');

INSERT INTO CONTRACT

(contract_num, contract_desc, estimated_cost, actual_cost, date_started, date_ended,
company_name)

VALUES

('C-56789', 'Short-term', 12000, 20000, '12-09-24', '28-10-24', 'Ford Harrison LTD');

INSERT INTO CONTRACT

(contract_num, contract_desc, estimated_cost, actual_cost, date_started, date_ended,
company_name)

VALUES

('C-12134', NULL, 59000, 59000, '20-10-24', NULL, 'Noel Road LTD');

COMMIT;

	CONTRACT_NUM	CONTRACT_DESC	ESTIMATED_COST	ACTUAL_COST	DATE_STARTED	DATE_ENDED	COMPANY_NAME
1	C-78643	(null)	20000	40000	11/09/24	22/10/24	Noel Road LTD
2	C-12134	(null)	59000	59000	20/10/24	(null)	Noel Road LTD
3	C-90967	Medium-term	120000	90000	20/10/24	(null)	Ford Harrison LTD
4	C-12356	Short-term	36000	42000	10/09/24	01/10/24	Buggy Machinery Roder
5	C-86064	Long-term	450000	200000	08/11/24	20/01/25	Roder Fixer LTD
6	C-12563	Long-term	786000	740000	20/10/24	26/10/24	Harvey Construction
7	C-56789	Short-term	12000	20000	12/09/24	28/10/24	Ford Harrison LTD

(new)

INSERT INTO CONTRACT

(contract_num, contract_name ,contract_desc, estimated_cost, actual_cost,
date_started, date_ended, company_name)

VALUES ('C-90967','Ford Contract', 'Medium-term', 120000, 90000, '20-10-24', NULL, 'Ford
Harrison LTD');

INSERT INTO CONTRACT

(contract_num, contract_name ,contract_desc, estimated_cost, actual_cost,
date_started, date_ended, company_name)

VALUES ('C-12356','Buggy Contract Agreement', 'Short-term', 36000, 42000, '10-09-24',
'01-10-24', 'Buggy Machinery Roder');

INSERT INTO CONTRACT

(contract_num, contract_name ,contract_desc, estimated_cost, actual_cost,
date_started, date_ended, company_name)

VALUES ('C-86064','Roder Contract', 'Long-term', 450000, 200000, '08-11-24', '20-01-25',
'Roder Fixer LTD');

INSERT INTO CONTRACT

```
(contract_num, contract_name , contract_desc, estimated_cost, actual_cost,
date_started, date_ended, company_name)
VALUES ('C-12563','Harvey Road Repair agreement', 'Long-term', 786000, 740000,
'20-10-24', '26-10-24', 'Harvey Construction');
```

INSERT INTO CONTRACT

```
(contract_num, contract_name, contract_desc, estimated_cost, actual_cost,
date_started, date_ended, company_name)
VALUES
('C-78643','Noel Road Contract', NULL, 20000, 40000, '11-09-24', '22-10-24', 'Noel Road LTD');
```

INSERT INTO CONTRACT

```
(contract_num, contract_name, contract_desc, estimated_cost, actual_cost,
date_started, date_ended, company_name)
VALUES ('C-56789','Ford Contract 2', 'Short-term', 12000, 20000, '12-09-24', '28-10-24',
'Ford Harrison LTD');
```

INSERT INTO CONTRACT

```
(contract_num, contract_name, contract_desc, estimated_cost, actual_cost,
date_started, date_ended, company_name)
VALUES ('C-12134','Noel Road Contract Agreement', NULL, 59000, 59000, '20-10-24',
NULL, 'Noel Road LTD');
```

COMMIT;

CONTRACT_NUM	CONTRACT_DESC	ESTIMATED_COST	ACTUAL_COST	DATE_STARTED	DATE_ENDED	COMPANY_NAME	CONTRACT_NAME
1 C-90967	Medium-term	120000	90000	20/10/24	(null)	Ford Harrison LTD	Ford Contract
2 C-12356	Short-term	36000	42000	10/09/24	01/10/24	Buggy Machinery Roder	Buggy Contract Agreement
3 C-86064	Long-term	450000	200000	08/11/24	20/01/25	Roder Fixer LTD	Roder Contract
4 C-12563	Long-term	786000	740000	20/10/24	26/10/24	Harvey Construction	Harvey Road Repair agreement
5 C-78643	(null)	20000	40000	11/09/24	22/10/24	Noel Road LTD	Noel Road Contract
6 C-56789	Short-term	12000	20000	12/09/24	28/10/24	Ford Harrison LTD	Ford Contract 2
7 C-12134	(null)	59000	59000	20/10/24	(null)	Noel Road LTD	Noel Road Contract Agreement

Entry Contractor

INSERT INTO CONTRACTOR

```
(company_name, company_address, contactdetails)
VALUES
('Ford Harrison LTD', 'Saint Georges Road', 0788956431);
```

INSERT INTO CONTRACTOR

```
(company_name, company_address, contactdetails)
VALUES
('Roder Fixer LTD', '55 Wellesley Street East', 0129876409);
```

```
INSERT INTO CONTRACTOR
(company_name, company_address, contactdetails)
VALUES
('Harvey Construction', '67 Crambold Street', 0567894223);
```

```
INSERT INTO CONTRACTOR
(company_name, company_address, contactdetails)
VALUES
('Noel Road LTD', 'Ham Ridge Road', 0226578943);
```

```
INSERT INTO CONTRACTOR
(company_name, company_address, contactdetails)
VALUES
('Buggy Machinery Roder', '90A Quarry St East', 0227654386);
```

```
COMMIT;
```

	COMPANY_NAME	COMPANY_ADDRESS	CONTACTDETAILS
1	Ford Harrison LTD	Saint Georges Road	788956431
2	Roder Fixer LTD	55 Wellesley Street East	129876409
3	Harvey Construction	67 Crambold Street	567894223
4	Noel Road LTD	Ham Ridge Road	226578943
5	Buggy Machinery Roder	90A Quarry St East	227654386

Entry RoadCategory

```
INSERT INTO ROADCATEGORY
(road_id, category_id)
VALUES
('R-4080', 02);
```

```
INSERT INTO ROADCATEGORY
(road_id, category_id)
VALUES
('R-4081', 02);
```

```
INSERT INTO ROADCATEGORY
(road_id, category_id)
VALUES
('R-4082', 02);
```

```
INSERT INTO ROADCATEGORY
```

```
(road_id, category_id)
VALUES
('R-1156', 04);
```

```
INSERT INTO ROADCATEGORY
(road_id, category_id)
VALUES
('R-3560', 01);
```

```
INSERT INTO ROADCATEGORY
(road_id, category_id)
VALUES
('R-3561', 01);
```

```
INSERT INTO ROADCATEGORY
(road_id, category_id)
VALUES
('R-8150', 02);
```

```
INSERT INTO ROADCATEGORY
(road_id, category_id)
VALUES
('R-5910', 03);
```

```
INSERT INTO ROADCATEGORY
(road_id, category_id)
VALUES
('R-5911', 03);
```

```
INSERT INTO ROADCATEGORY
(road_id, category_id)
VALUES
('R-3450', 05);
COMMIT;
```

	ROAD_ID	CATEGORY_ID
1	R-1156	4
2	R-3450	5
3	R-3560	1
4	R-3561	1
5	R-4080	2
6	R-4081	2
7	R-4082	2
8	R-5910	3
9	R-5911	3
10	R-8150	2

Entry StartEnd

<new code>

```
INSERT INTO STARTEND (LOCATION_ID, ROAD_ID, LOCATIONSTART, LOCATIONEND)
VALUES ('LOC-40', 'R-4080', '1 Bucklands Beach', '35 Bucklands Beach');
```

```
INSERT INTO STARTEND (LOCATION_ID, ROAD_ID, LOCATIONSTART, LOCATIONEND)
VALUES ('LOC-41', 'R-4081', '1 Buck Beach', '44 Buck Beach');
```

```
INSERT INTO STARTEND (LOCATION_ID, ROAD_ID, LOCATIONSTART, LOCATIONEND)
VALUES ('LOC-42', 'R-4082', '1 Sunderland Beach', '50 Sunderland Beach');
```

```
INSERT INTO STARTEND (LOCATION_ID, ROAD_ID, LOCATIONSTART, LOCATIONEND)
VALUES ('LOC-15', 'R-1156', '1 Elstree Avenue', '20 Elstree Avenue');
```

```
INSERT INTO STARTEND (LOCATION_ID, ROAD_ID, LOCATIONSTART, LOCATIONEND)
VALUES ('LOC-30', 'R-3560', 'State Highway 20', 'State Highway 20B');
```

```
INSERT INTO STARTEND (LOCATION_ID, ROAD_ID, LOCATIONSTART, LOCATIONEND)
VALUES ('LOC-31', 'R-3561', 'South-Western 12', 'South-Western 12B');
```

```
INSERT INTO STARTEND (LOCATION_ID, ROAD_ID, LOCATIONSTART, LOCATIONEND)
VALUES ('LOC-80', 'R-8150', '1 Great South', '101 Great South');
```

```
INSERT INTO STARTEND (LOCATION_ID, ROAD_ID, LOCATIONSTART, LOCATIONEND)
VALUES ('LOC-10', 'R-5910', '1 Panama Road', '25 Panama Road');
```

```
INSERT INTO STARTEND (LOCATION_ID, ROAD_ID, LOCATIONSTART, LOCATIONEND)
VALUES ('LOC-11', 'R-5911', '1 McLennan Road', '40 McLennan Road');
```

```
INSERT INTO STARTEND (LOCATION_ID, ROAD_ID, LOCATIONSTART, LOCATIONEND)
VALUES ('LOC-34', 'R-3450', '1 Te Kehu Way', '24 Te Kehu Way');
COMMIT;
```

	LOCATION_ID	ROAD_ID	LOCATIONSTART	LOCATIONEND
1	LOC-40	R-4080	1 Bucklands Beach	35 Bucklands Beach
2	LOC-41	R-4081	1 Buck Beach	44 Buck Beach
3	LOC-42	R-4082	1 Sunderland Beach	50 Sunderland Beach
4	LOC-15	R-1156	1 Elstree Avenue	20 Elstree Avenue
5	LOC-30	R-3560	State Highway 20	State Highway 20B
6	LOC-31	R-3561	South-Western 12	South-Western 12B
7	LOC-80	R-8150	1 Great South	101 Great South
8	LOC-10	R-5910	1 Panama Road	25 Panama Road
9	LOC-11	R-5911	1 McLennan Road	40 McLennan Road
10	LOC-34	R-3450	1 Te Kehu Way	24 Te Kehu Way

Old code

```
INSERT INTO STARTEND
(location_id, road_id, locationstart, locationend,)
VALUES
('R-4080', 'LOC-40', '1 Bucklands Beach', '35 Buckland Beach');
```

```
INSERT INTO STARTEND
(location_id, road_id, locationstart, locationend)
VALUES
('R-4080', 'LOC-40', '1 Bucklands Beach', '35 Bucklands Beach');
```

```
INSERT INTO STARTEND
(location_id, road_id, locationstart, locationend)
VALUES
('R-4081', 'LOC-41', '1 Buck Beach', '44 Buck Beach');
```

```
INSERT INTO STARTEND
(location_id, road_id, locationstart, locationend)
```

```
VALUES  
( 'R-4082', 'LOC-42', '1 Sunderland Beach', '50 Sunderland Beach');
```

```
INSERT INTO STARTEND  
(location_id, road_id, locationstart, locationend)  
VALUES  
( 'R-1156', 'LOC-15', '1 Elstree Avenue', '20 Elstree Avenue');
```

```
INSERT INTO STARTEND  
(location_id, road_id, locationstart, locationend)  
VALUES  
( 'R-3560', 'LOC-30', 'State Highway 20', 'State Highway 20B');
```

```
INSERT INTO STARTEND  
(location_id, road_id, locationstart, locationend)  
VALUES  
( 'R-3561', 'LOC-31', 'South-Western 12', 'South-Western 12B');
```

```
INSERT INTO STARTEND  
(location_id, road_id, locationstart, locationend)  
VALUES  
( 'R-8150', 'LOC-80', '1 Great South', '101 Great South');
```

```
INSERT INTO STARTEND  
(location_id, road_id, locationstart, locationend)  
VALUES  
( 'R-5910', 'LOC-10', '1 Panama Road', '25 Panama Road');
```

```
INSERT INTO STARTEND  
(location_id, road_id, locationstart, locationend)  
VALUES  
( 'R-5911', 'LOC-11', '1 McLennan Road', '40 McLennan Road');
```

```
INSERT INTO STARTEND  
(location_id, road_id, locationstart, locationend)  
VALUES  
( 'R-3450', 'LOC-34', '1 Te Kehu Way', '24 Te Kehu Way');  
COMMIT;
```

Entry Record

```
INSERT INTO RECORD(project_code, road_id)  
VALUES ('F4KJ3Q', 'R-4080');
```

```
INSERT INTO RECORD (project_code, road_id)
VALUES ('F4KJ3Q', 'R-4081');
```

```
INSERT INTO RECORD (project_code, road_id)
VALUES ('NVJWQR', 'R-4082');
```

```
INSERT INTO RECORD (project_code, road_id)
VALUES ('P4ETHJ', 'R-1156');
```

```
INSERT INTO RECORD (project_code, road_id)
VALUES ('4ZCG7Z', 'R-3560');
```

```
INSERT INTO RECORD (project_code, road_id)
VALUES ('4ZCG7Z', 'R-3561');
```

```
INSERT INTO RECORD (project_code, road_id)
VALUES ('GKFVAQ', 'R-8150');
```

```
INSERT INTO RECORD (project_code, road_id)
VALUES ('TYQ8AQ', 'R-5910');
```

```
INSERT INTO RECORD (project_code, road_id)
VALUES ('TYQ8AQ', 'R-5911');
```

```
INSERT INTO RECORD (project_code, road_id)
VALUES ('AB7A3M', 'R-3450');
```

```
COMMIT;
```


	PROJECT_CODE	RO...	
1	4ZCG7Z	R-3560	
2	4ZCG7Z	R-3561	
3	AB7A3M	R-3450	
4	F4KJ3Q	R-4080	
5	F4KJ3Q	R-4081	
6	GKFVAQ	R-8150	
7	NVJWQR	R-4082	
8	P4ETHJ	R-1156	
9	TYQ8AQ	R-5910	
10	TYQ8AQ	R-5911	

Entry Staff_record

<new code>

```
INSERT INTO STAFF_RECORD (PROJECT_CODE, EMPLOYEE_ID)
VALUES ('F4KJ3Q', 'EP-001');
```

```
INSERT INTO STAFF_RECORD (PROJECT_CODE, EMPLOYEE_ID)
VALUES ('F4KJ3Q', 'EP-002');
```

```
INSERT INTO STAFF_RECORD (PROJECT_CODE, EMPLOYEE_ID)
VALUES ('F4KJ3Q', 'EP-101');
```

```
INSERT INTO STAFF_RECORD (PROJECT_CODE, EMPLOYEE_ID)
VALUES ('NVJWQR', 'EP-201');
```

```
INSERT INTO STAFF_RECORD (PROJECT_CODE, EMPLOYEE_ID)
VALUES ('NVJWQR', 'EP-202');
```

```
INSERT INTO STAFF_RECORD (PROJECT_CODE, EMPLOYEE_ID)
VALUES ('4ZCG7Z', 'EP-203');
```

```
INSERT INTO STAFF_RECORD (PROJECT_CODE, EMPLOYEE_ID)
VALUES ('P4ETHJ', 'EP-021');
```

```
INSERT INTO STAFF_RECORD (PROJECT_CODE, EMPLOYEE_ID)
VALUES ('GKFVAQ', 'EP-221');
```

```
INSERT INTO STAFF_RECORD(PROJECT_CODE, EMPLOYEE_ID)
VALUES('P4ETHJ', 'EP-081');
```

```
INSERT INTO STAFF_RECORD(PROJECT_CODE, EMPLOYEE_ID)
VALUES('4ZCG7Z', 'EP-012');
```

```
INSERT INTO STAFF_RECORD(PROJECT_CODE, EMPLOYEE_ID)
VALUES('GKFVAQ', 'EP-110');
```

```
INSERT INTO STAFF_RECORD(PROJECT_CODE, EMPLOYEE_ID)
VALUES('TYQ8AQ', 'EP-021');
```

```
INSERT INTO STAFF_RECORD(PROJECT_CODE, EMPLOYEE_ID)
VALUES('AB7A3M', 'EP-134');
```

```
INSERT INTO STAFF_RECORD(PROJECT_CODE, EMPLOYEE_ID)
VALUES('AB7A3M', 'EP-122');
COMMIT;
```

186	SELECT *	FROM STAFF_RECORD;
187		
188		

Script Output x	Autotrace x	Query Result x	▶ Qu
SQL	All Rows Fetched: 14 in 0.012 seconds		

	PROJECT_CODE	EMPLOYEE_ID
1	4ZCG7Z	EP-012
2	4ZCG7Z	EP-203
3	AB7A3M	EP-122
4	AB7A3M	EP-134
5	F4KJ3Q	EP-001
6	F4KJ3Q	EP-002
7	F4KJ3Q	EP-101
8	GKFVAQ	EP-110
9	GKFVAQ	EP-221
10	NVJWQR	EP-201
11	NVJWQR	EP-202
12	P4ETHJ	EP-021
13	P4ETHJ	EP-081
14	TYQ8AQ	EP-025

OLD CODE

```
INSERT INTO STAFF_RECORD
(project_code, employee_id)
VALUES
('F4KJ3Q', 'EP-001');
```

```
INSERT INTO STAFF_RECORD
(project_code, employee_id)
VALUES
('F4KJ3Q', 'EP-001');
```

```
INSERT INTO STAFF_RECORD  
(project_code, employee_id)  
VALUES  
('F4KJ3Q', 'EP-002');
```

```
INSERT INTO STAFF_RECORD  
(project_code, employee_id)  
VALUES  
('F4KJ3Q', 'EP-101');
```

```
INSERT INTO STAFF_RECORD  
(project_code, employee_id)  
VALUES  
('NVJWQR', 'EP-201');
```

```
INSERT INTO STAFF_RECORD  
(project_code, employee_id)  
VALUES  
('NVJWQR', 'EP-202');
```

```
INSERT INTO STAFF_RECORD  
(project_code, employee_id)  
VALUES  
('4ZCG7Z', 'EP-203');
```

```
INSERT INTO STAFF_RECORD  
(project_code, employee_id)  
VALUES  
('P4ETHJ', 'EP-021');
```

```
INSERT INTO STAFF_RECORD  
(project_code, employee_id)  
VALUES  
('GKFVAQ', 'EP-221');
```

```
INSERT INTO STAFF_RECORD  
(project_code, employee_id)  
VALUES  
('P4ETHJ', 'EP-081');
```

```
INSERT INTO STAFF_RECORD  
(project_code, employee_id)  
VALUES
```

```
('4ZCG7Z', 'EP-012');
```

```
INSERT INTO STAFF_RECORD  
(project_code, employee_id)  
VALUES  
('GKFVAQ', 'EP-110');
```

```
INSERT INTO STAFF_RECORD  
(project_code, employee_id)  
VALUES  
('TYQ8AQ', 'EP-021');
```

```
INSERT INTO STAFF_RECORD  
(project_code, employee_id)  
VALUES  
('AB7A3M', 'EP-134');
```

```
INSERT INTO STAFF_RECORD  
(project_code, employee_id)  
VALUES  
('AB7A3M', 'EP-122');
```

```
COMMIT;
```

Entry Staffrole

```
INSERT INTO STAFFROLE  
(role_name, employee_id, assigned_date, assigned_enddate)  
VALUES  
('Road Leader', 'EP-001', '20-09-22', '16-08-23');
```

```
INSERT INTO STAFFROLE  
(role_name, employee_id, assigned_date, assigned_enddate)  
VALUES  
('Manager', 'EP-001', '16-08-23', NULL);
```

```
INSERT INTO STAFFROLE  
(role_name, employee_id, assigned_date, assigned_enddate)  
VALUES  
('Road Worker', 'EP-002', '20-09-22', '14-07-23');
```

```
INSERT INTO STAFFROLE  
(role_name, employee_id, assigned_date, assigned_enddate)
```

VALUES
('Road Machine Operator', 'EP-002', '14-07-23', NULL);

INSERT INTO STAFFROLE
(role_name, employee_id, assigned_date, assigned_enddate)
VALUES
('Road Machine Operator', 'EP-101', '12-01-23', NULL);

INSERT INTO STAFFROLE
(role_name, employee_id, assigned_date, assigned_enddate)
VALUES
('Road Worker', 'EP-201', '15-09-23', '20-12-23');

INSERT INTO STAFFROLE
(role_name, employee_id, assigned_date, assigned_enddate)
VALUES
('Road Machine Operator', 'EP-201', '20-12-23', NULL);

INSERT INTO STAFFROLE
(role_name, employee_id, assigned_date, assigned_enddate)
VALUES
('Road Worker', 'EP-202', '20-09-23', NULL);

INSERT INTO STAFFROLE
(role_name, employee_id, assigned_date, assigned_enddate)
VALUES
('Road Worker', 'EP-203', '20-09-23', NULL);

INSERT INTO STAFFROLE
(role_name, employee_id, assigned_date, assigned_enddate)
VALUES
('Manager', 'EP-021', '30-11-22', '16-08-23');

INSERT INTO STAFFROLE
(role_name, employee_id, assigned_date, assigned_enddate)
VALUES
('Road Leader', 'EP-021', '16-08-23', NULL);

INSERT INTO STAFFROLE
(role_name, employee_id, assigned_date, assigned_enddate)
VALUES
('Road Worker', 'EP-221', '4-11-23', NULL);

INSERT INTO STAFFROLE

```
(role_name, employee_id, assigned_date, assigned_enddate)
VALUES
('Road Worker', 'EP-081', '15-12-22', NULL);
```

```
INSERT INTO STAFFROLE
(role_name, employee_id, assigned_date, assigned_enddate)
VALUES
('Road Worker', 'EP-012', '26-11-22', NULL);
```

```
INSERT INTO STAFFROLE
(role_name, employee_id, assigned_date, assigned_enddate)
VALUES
('Road Traffic Controller', 'EP-110', '12-08-22', NULL);
```

```
INSERT INTO STAFFROLE
(role_name, employee_id, assigned_date, assigned_enddate)
VALUES
('Road Traffic Controller', 'EP-021', '10-06-22', NULL);
```

```
INSERT INTO STAFFROLE
(role_name, employee_id, assigned_date, assigned_enddate)
VALUES
('Road Traffic Controller', 'EP-134', '23-07-23', NULL);
```

```
INSERT INTO STAFFROLE
(role_name, employee_id, assigned_date, assigned_enddate)
VALUES
('Road Traffic Controller', 'EP-122', '12-05-22', NULL);
COMMIT;
```

	ROLE_NAME	EMPLOYEE_ID	ASSIGNED_DATE	ASSIGNED_ENDDATE
1	Road Leader	EP-001	20/09/22	16/08/23
2	Manager	EP-001	16/08/23	(null)
3	Road Worker	EP-002	20/09/22	14/07/23
4	Road Machine Operator	EP-002	14/07/23	(null)
5	Road Machine Operator	EP-101	12/01/23	(null)
6	Road Worker	EP-201	15/09/23	20/12/23
7	Road Machine Operator	EP-201	20/12/23	(null)
8	Road Worker	EP-202	20/09/23	(null)
9	Road Worker	EP-203	20/09/23	(null)
10	Road Leader	EP-021	16/08/23	(null)
11	Road Worker	EP-221	04/11/23	(null)
12	Road Worker	EP-081	15/12/22	(null)
13	Road Worker	EP-012	26/11/22	(null)
14	Road Traffic Controller	EP-110	12/08/22	(null)
15	Road Traffic Controller	EP-021	10/06/22	(null)
16	Road Traffic Controller	EP-134	23/07/23	(null)
17	Road Traffic Controller	EP-122	12/05/22	(null)
18	Manager	EP-021	30/11/22	16/08/23

Assumptions

A road can be counted as multiple categories (unsealed road,urban road, residential, dirt road).

A road can be a subsection of multiple roads or not named road.

- (new) A road may have many subsections

A subsection may be part of one road

A role must have one and have multiple contracts.

The sub_section value correlate to road_id value if there is a subsection for the road.

Assigned end dates are the last day one staff is involved in a project therefore it can be nulled if the project is unfinished.

A location must be assigned to many StartEnd or else some road will not have a start location or end location

A Contract may have many contract managers and StaffRole can only assign one manager to a Contract

Assumptions

1. A road can be a subsection of multiple roads or not named road.
2. In a contract there can be multiple roles
- 3.
- 4.
- 5.

SQL QUERY CODE:

Sql query code 1

Purpose of the query:

This query is to search for which staff is assigned to a specific project (e.g.) and what contract that project is assigned to. This will retrieve the employee ID, employee name (concatenation of last and first name), project code and contract number. This will be ordered by project code.

SQL SELECT Query

```
SELECT s.employee_id AS "Employee ID", s.firstname||' '||s.lastname AS "Employee Name",  
p.project_code AS "Project Code", c.contract_num AS "Contract Number"  
FROM staff s, project p, contract c, staff_record r  
WHERE s.employee_id = r.employee_id  
AND p.project_code = r.project_code  
AND p.contract_num = c.contract_num  
ORDER BY p.project_code;
```

Output/Result of the query

	Employee ID	Employee Name	Project Code	Contract Number
1	EP-203	Toast Birch	4ZCG7Z	C-12563
2	EP-012	Arnold Swan	4ZCG7Z	C-12563
3	EP-122	Anderi Zeph	AB7A3M	C-12134
4	EP-134	Migel Santos	AB7A3M	C-12134
5	EP-002	Welis Arow	F4KJ3Q	C-90967
6	EP-001	Bob Forest	F4KJ3Q	C-90967
7	EP-101	Carl Waters	F4KJ3Q	C-90967
8	EP-110	Roger Mary	GKFVAQ	C-78643
9	EP-221	Crane Gunn	GKFVAQ	C-78643
10	EP-202	Ale Birch	NVJWQR	C-12356
11	EP-201	Jim Karl	NVJWQR	C-12356
12	EP-081	York Joseph	P4ETHJ	C-86064
13	EP-021	Beth Sun	P4ETHJ	C-86064
14	EP-025	Mike Hunt	TYQ8AQ	C-56789

SQL QUERY CODE 2

Purpose of the query:

This query purpose is to search for information about sub section roads, specifically when a subsection road ends and starts as well as its coordinates, sub-section and road category. This will retrieve the Road name, Road length, Location End, Location Start, Coordinates(concatenation of latitude and longitude), sub_section and Category name. Ordered by subsection

SQL SELECT QUERY:

```
SELECT r.road_name AS "Road Name", r.road_length AS "Road Length",
se.locationstart AS "Start", se.locationend AS "End",
l.latitude||', '||l.longitude AS "Coordinates", r2.sub_section AS "Sub-Section"
, c.category_name AS "Category Type"
FROM road r, startend se, location l, roadcategory rc, category c, road r2
WHERE r.road_id = se.road_id
AND l.location_id = se.location_id
AND r.road_id = rc.road_id
AND c.category_id = rc.category_id
AND r.road_id = r2.road_id
AND r2.sub_section LIKE 'R%'
ORDER BY r2.sub_section;
```

Output/Result of the query

Script Output x Query Result x

 | All Rows Fetched: 4 in 0.019 seconds

ROAD_NAME	ROAD_LENGTH	LOCATIONSTART	LOCATIONEND	L.LATITUDE	L.LONGITUDE	SUB_SECTION	CATEGORY_NAME
1 South-Western 12	12	South-Western 12	South-Western 12B	40-154		R-3560	Main Highway
2 Buck Beach	5	1 Buck Beach	44 Buck Beach	-36156		R-4080	Road
3 Sunderland Beach	5	1 Sunderland Beach	50 Sunderland Beach	-36140		R-4080	Road
4 McLennan Road	4	1 McLennan Road	40 McLennan Road	-24412		R-5910	Secondary road

SQL QUERY CODE 3

Purpose of the query:

The purpose of this query is to search for the list of roads that are currently or were once in repair with an actual cost greater than \$40,000 and searching what projects, contracts and contractors are assigned to these roads. This will retrieve Road name, Project name, Contract name, Company name and Actual cost. Ordered by Actual cost.

SQL SELECT QUERY:

```
SELECT r.road_name AS "Road Name", p.project_name AS "Project Name",
c.contract_name AS "Contract_name", c.actual_cost AS "Actual Cost"
FROM road r, record rd, project p, contract c, contractor cr
WHERE r.road_id = rd.road_id
AND p.project_code = rd.project_code
AND p.contract_num = c.contract_num
AND c.company_name = cr.company_name
GROUP BY r.road_name, p.project_name, c.contract_name, c.actual_cost
HAVING actual_cost > 40000
ORDER BY actual_cost DESC;
```

Output/Result of the query

ROAD_NAME	PROJECT_NAME	CONTRACT_NAME	ACTUAL_COST
1 South-Western 12	Project Gunn	Harvey Road Repair agreement	740000
2 State Highway 20	Project Gunn	Harvey Road Repair agreement	740000
3 Elstree Avenue	Evergreen	Roader Contract	200000
4 Buck Beach	Melted Butter	Ford Contract	90000
5 Bucklands Beach	Melted Butter	Ford Contract	90000
6 Te Kehu Way	Waterflow	Noel Road Contract Agreement	59000
7 Sunderland Beach	Smooth rock	Buggy Contract Agreement	42000

SQL QUERY CODE 4:

Purpose of the query: The purpose of this query is to find staff and their specific role that still have ongoing projects. This query can also be used by the manager to manage assigned dates and end dates. This query includes Employee ID, First name, Last name, Phone number, Role name, Assigned end date.

SQL SELECT QUERY:

```
SELECT s.employee_id AS "Employee ID", s.firstname AS "First Name", s.lastname AS "Last Name",  
s.phone_num AS "Phone Number", r.role_name AS "Role Name",  
sr.assigned_enddate AS "Assigned End Date"  
FROM staff s  
JOIN staffrole sr ON s.employee_id = sr.employee_id  
JOIN role r ON sr.role_name = r.role_name  
WHERE sr.assigned_enddate IS NULL  
ORDER BY s.employee_id;
```

Output/Result of the query

	Employee ID	First Name	Last Name	Phone Number	Role Name	Assigned End Date
1	EP-001	Bob	Forest	924797635	Manager	(null)
2	EP-002	Welis	Arow	273748284	Road Machine Operator	(null)
3	EP-012	Arnold	Swan	647548476	Road Worker	(null)
4	EP-021	Beth	Sun	766546887	Road Leader	(null)
5	EP-021	Beth	Sun	766546887	Road Traffic Controller	(null)
6	EP-081	York	Joseph	123456782	Road Worker	(null)
7	EP-101	Carl	Waters	776439439	Road Machine Operator	(null)
8	EP-110	Roger	Mary	912649154	Road Traffic Controller	(null)
9	EP-122	Anderi	Zeph	765926541	Road Traffic Controller	(null)
10	EP-134	Migel	Santos	981490186	Road Traffic Controller	(null)
11	EP-201	Jim	Karl	355772984	Road Machine Operator	(null)
12	EP-202	Ale	Birch	822935233	Road Worker	(null)
13	EP-203	Toast	Birch	988569987	Road Worker	(null)
14	EP-221	Crane	Gunn	234567876	Road Worker	(null)

SQL QUERY CODE 5

Purpose of the query: The purpose of this query is to find the record of difference between actual cost and estimated cost of a project from the contracts table that has a difference under zero. This query includes Project name, Estimated cost, Actual cost.

SQL SELECT QUERY:

```
SELECT p.project_name AS "Project Name",
TO_CHAR(c.estimated_cost, '$999,999.00') AS "Estimated Cost",
TO_CHAR(c.actual_cost, '$999,999.00') AS "Actual Cost",
'||r.road_name AS "Road Name",
TO_CHAR(c.actual_cost - c.estimated_cost, '$999,999.00') AS "Cost Difference"
FROM project p, contract c, road r, record rd
WHERE p.contract_num = c.contract_num
AND r.road_id = rd.road_id
AND p.project_code = rd.project_code
GROUP BY p.project_name, c.estimated_cost, c.actual_cost , r.road_name,
(actual_cost - estimated_cost)
HAVING (actual_cost - estimated_cost) < 0
ORDER BY (actual_cost - estimated_cost)DESC;
```

Output/Result of the query

	Project Name	Estimated Cost	Actual Cost	Road Name	Cost Difference
1	Melted Butter	\$120,000.00	\$90,000.00	Buck Beach	-\$30,000.00
2	Melted Butter	\$120,000.00	\$90,000.00	Bucklands Beach	-\$30,000.00
3	Project Gunn	\$786,000.00	\$740,000.00	South-Western 12	-\$46,000.00
4	Project Gunn	\$786,000.00	\$740,000.00	State Highway 20	-\$46,000.00
5	Evergreen	\$450,000.00	\$200,000.00	Elstree Avenue	-\$250,000.00

